

William Anderson

Senior Software Engineer

Seminole, FL | williamandresondev@outlook.com | +1 (407) 801-1287 |
<https://www.linkedin.com/in/william-andreson-39b88b3aa>

Professional Summary

Senior Software Engineer with 12+ years building SaaS, healthcare, finance, e-commerce, and enterprise workflow platforms using **Ruby on Rails 4.x–7.x**, **React 16.x–18.x**, **TypeScript 3.x–5.x**, **PostgreSQL 9.x–15.x**, **Redis 4.x–7.x**, **Sidekiq 5.x–7.x**, **GraphQL**, **REST APIs**, and **AWS**, specializing in modernizing legacy Rails monoliths, React component architecture, secure API design, performance tuning, CI/CD automation, production debugging, and measurable delivery where upgrades, bug fixes, and new features improved reliability, reduced page load time, and helped product teams ship faster across regulated and customer-facing environments.

Technical Skills

- **Languages:** Ruby 2.x–3.x, JavaScript ES6+, TypeScript 3.x–5.x, SQL, HTML5, CSS3, Bash
- **Backend:** **Ruby on Rails 4.x–7.x**, Rails API Mode, ActiveRecord, ActionCable, Sidekiq, Redis, RSpec, Capybara, Puma, Rack, GraphQL, REST APIs
- **Frontend:** **React 16.x–18.x**, React Hooks, Redux, Context API, React Query, JavaScript ES6+, TypeScript, Webpack, Vite, Storybook, Material UI, Tailwind CSS
- **Database:** **PostgreSQL 9.x–15.x**, MySQL, Redis, query optimization, indexing, migrations, schema refactoring, data cleanup
- **Cloud & DevOps:** **AWS**, EC2, S3, RDS, ElastiCache, CloudWatch, Docker, GitHub Actions, GitLab CI, CircleCI, Nginx, Linux
- **Testing & Quality:** RSpec, Jest, React Testing Library, Cypress, Capybara, RuboCop, ESLint, Prettier, CI quality gates
- **Architecture:** Monolith modernization, service extraction, event-driven jobs, API-first design, multi-tenant SaaS, caching, background processing

Professional Experience

Viaro Networks Inc.

Senior Software Engineer | *January 2021 – December 2025*

- Architected customer-facing SaaS modules using **Ruby on Rails 6.x–7.x** and **React 17.x–18.x**, improving workflow speed by **38%** through cleaner API boundaries, reusable UI components, and optimized database access.
- Migrated a legacy Rails application from **Rails 5.x to Rails 7.x** and Ruby from **2.6 to 3.2**, resolving dependency conflicts, Zeitwerk autoloading errors, broken gems, and production deployment issues without major downtime.
- Built secure REST and **GraphQL APIs** for account management, billing, reporting, and admin workflows, reducing frontend/backend integration defects by **31%** through contract-based validation and shared response patterns.
- Implemented a React dashboard with **TypeScript 4.x–5.x**, React Query, and reusable form components, cutting duplicate frontend code by **42%** and improving feature delivery consistency across product teams.

- Resolved slow PostgreSQL queries caused by missing compound indexes, inefficient ActiveRecord joins, and N+1 loading, reducing several high-traffic page response times from seconds to sub-second averages.
- Refactored background processing with **Sidekiq 6.x–7.x** and Redis queues, fixing retry storms, stuck jobs, and race conditions that previously caused delayed notifications and inconsistent report generation.
- Designed role-based access control across Rails policies, API guards, and React route permissions, strengthening customer data isolation for enterprise users and regulated business workflows.
- Introduced **RSpec**, Jest, React Testing Library, and CI quality checks into older modules, raising automated coverage by **27%** while reducing regression bugs during monthly production releases.
- Improved AWS deployment stability by standardizing Docker builds, GitHub Actions pipelines, RDS migration steps, and CloudWatch alerts for Rails, Sidekiq, and React asset delivery.
- Fixed a recurring production memory leak caused by large ActiveRecord collections and unbounded JSON serialization, replacing eager loading with pagination, batching, and serializer-level controls.
- Delivered new real-time activity features using Rails ActionCable, Redis, and React state synchronization, giving operations teams faster visibility into customer events and system status.
- Mentored engineers on **Rails 7.x**, React Hooks, API design, test strategy, and debugging habits, helping the team review pull requests faster and reduce repeated architectural mistakes.

VividCloud

Software Engineer | January 2016 – December 2020

- Developed multi-tenant web applications using **Ruby on Rails 5.x–6.x**, **React 16.x–17.x**, PostgreSQL, Redis, and Sidekiq for healthcare, logistics, and subscription-based SaaS clients.
- Migrated frontend screens from server-rendered ERB and jQuery flows into React components, improving UI maintainability by **36%** and reducing repeated form-validation issues across customer portals.
- Upgraded a core Rails platform from **Rails 4.2 to Rails 6.x**, resolving asset pipeline breakages, strong parameter inconsistencies, deprecated callbacks, and gem compatibility issues during the migration.
- Built billing, user-management, and analytics features with Rails APIs and React dashboards, allowing business users to track subscriptions, usage, and operational metrics from one interface.
- Fixed intermittent production errors caused by Sidekiq job duplication and non-idempotent payment retries, adding locking, audit records, and safe retry logic to protect customer billing workflows.
- Optimized PostgreSQL reporting tables with better indexes, materialized views, and ActiveRecord query cleanup, reducing heavy report generation time by **44%** for operations teams.
- Implemented React component patterns with Redux, controlled forms, and shared validation helpers, making complex onboarding flows easier to extend without breaking existing user journeys.
- Integrated third-party APIs for payments, email delivery, address verification, and document generation, adding circuit-breaker handling and structured logs for faster production troubleshooting.
- Improved release reliability with CircleCI, Dockerized test environments, RuboCop, ESLint, and automated migration checks before deploying Rails and React changes to staging.
- Resolved browser-specific React rendering bugs in Safari and older Chrome versions by replacing fragile DOM assumptions with stable component state and tested event handling.
- Collaborated with product managers, designers, and QA engineers to convert ambiguous business requirements into tested Rails models, API endpoints, React screens, and measurable delivery milestones.

Minimalist Moon

Software Developer | January 2013 – December 2015

- Built e-commerce and content-management features using **Ruby on Rails 4.x–5.x**, PostgreSQL, Redis, JavaScript ES6, React 0.13–15.x, and responsive frontend patterns for growing digital brands.
- Introduced early React components for product filters, cart interactions, and admin tools, reducing duplicated JavaScript behavior by **29%** while keeping Rails views stable for existing customers.

- Migrated selected legacy modules from Rails 3.x conventions into **Rails 4.x–5.x**, fixing routing changes, mass-assignment risks, asset issues, and outdated authentication logic.
- Implemented inventory, order-tracking, promotional-code, and customer-profile features with ActiveRecord models, service objects, and clean controller boundaries for maintainable business workflows.
- Fixed checkout bugs caused by stale cart sessions, concurrent inventory updates, and incomplete payment callbacks, adding transactional safeguards and clearer failure recovery paths.
- Improved PostgreSQL performance by replacing expensive wildcard searches, adding targeted indexes, and introducing Redis caching for frequently accessed catalog and promotion data.
- Built admin reporting screens with Rails, React, and chart components, giving non-technical teams faster access to sales, refund, and fulfillment metrics without manual spreadsheet work.
- Added RSpec request specs, model specs, and Capybara feature tests around critical purchase paths, reducing release defects by **24%** across checkout and account-management flows.
- Supported AWS-hosted Rails deployments by improving Nginx configuration, Puma settings, environment variables, and log visibility for production debugging.
- Worked flexibly across backend, frontend, database, and deployment tasks, turning loosely defined product requests into stable features that could be maintained by a small engineering team.

Microsoft

Software Developer Intern | July 2012 – December 2012

- Contributed to internal engineering tools using **Ruby 1.9–2.x**, **Rails 3.x**, jQuery, Backbone.js, SQL, and early API-driven frontend patterns for developer productivity workflows.
- Built small Rails-based dashboards for tracking internal requests, data cleanup tasks, and team workflow status, helping engineers reduce manual follow-up effort by **18%**.
- Fixed form validation bugs, broken JavaScript interactions, and inconsistent server-side error handling that caused confusing user experiences in internal workflow tools.
- Refactored older Rails controllers by moving repeated business logic into service-style classes, making code easier to test and safer for future engineers to extend.
- Created SQL queries and lightweight reporting views that helped teams inspect operational data, identify duplicate records, and resolve workflow bottlenecks faster.
- Added unit and integration tests around critical Rails models and controller actions, improving confidence before changes were merged into shared internal applications.
- Supported migration cleanup from older Ruby gem versions by identifying deprecated methods, updating test fixtures, and validating behavior in staging-like environments.
- Collaborated with senior engineers during code reviews, learning practical debugging, source-control discipline, secure coding habits, and production-minded software delivery.
- Documented setup steps, known issues, and troubleshooting notes for internal Rails tools, making onboarding smoother for new contributors and reducing repeated environment questions.

Education

Florida State University

Bachelor's Degree in Computer Science | 2008 – 2012